THE STORY OF ELECTRICITY

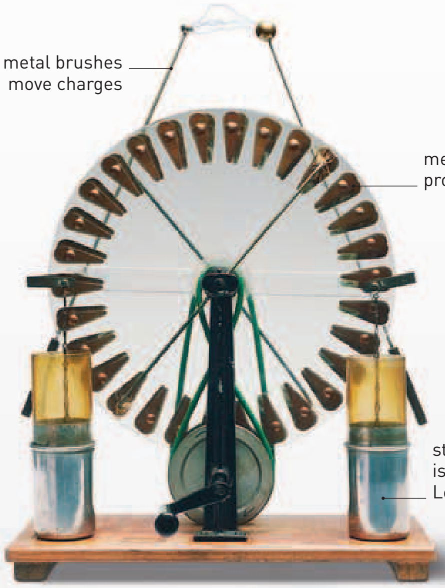
HARNESSING THE FORCE THAT BUILT THE MODERN WORLD

When Thales, a mathematician and philosopher in ancient Greece, experimented with a piece of amber—known as "elektron" in Greek—little could he have known that his initial observations would still hold a significant place in science more than 2,000 years later.

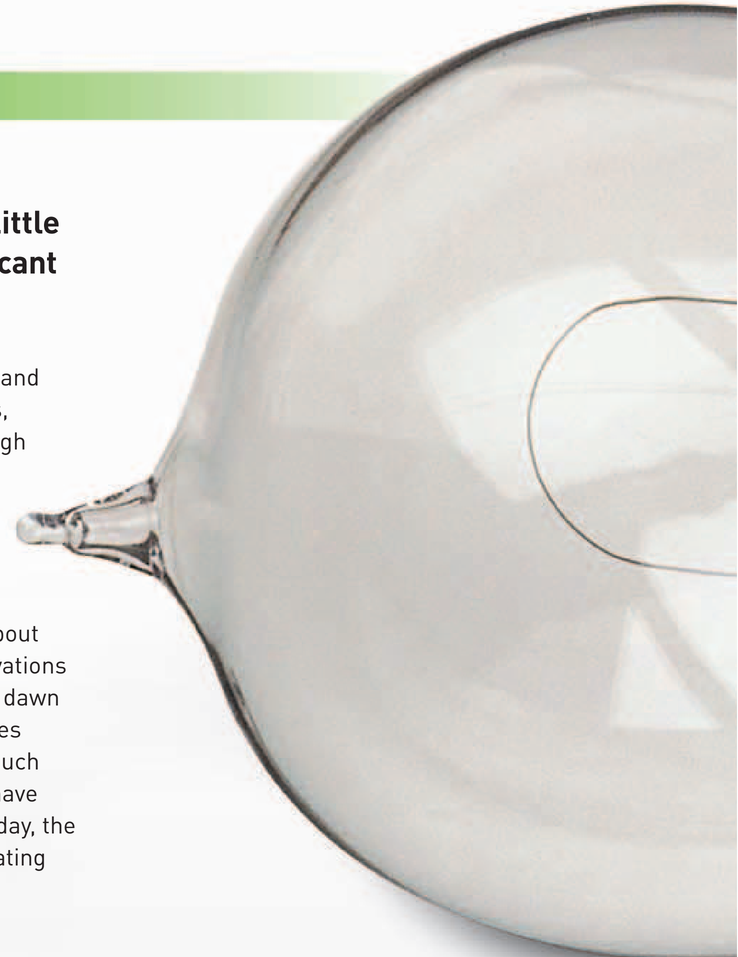
What Thales noticed was that if he rubbed a piece of amber against fur it would attract bits of dust and feathers lying nearby—although he did not know it, he had stumbled on what we know today as static electricity. Over the following centuries, scientists all over the world experimented with this form of electricity, as well as magnets and magnetism. By the 17th and 18th centuries, technological leaps had been made, although the connection between electricity and magnets would not be clear until the 19th century (see panel, right).

POWER TO THE PEOPLE

As the 1800s progressed, understanding about electricity rapidly increased, and new innovations were rolled out in quick succession. By the dawn of the 20th century, many of the technologies were in place that are still with us today—such as batteries and light bulbs—though they have since been further adapted and refined. Today, the scientific challenge is to find ways of generating electricity that do not cause pollution.

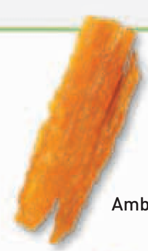


Wimshurst machine
The English inventor James Wimshurst developed a device that could generate static electricity and store it in a vessel called a Leyden jar. For many years, scientists studying electricity used Wimshurst machines to produce electric charge.

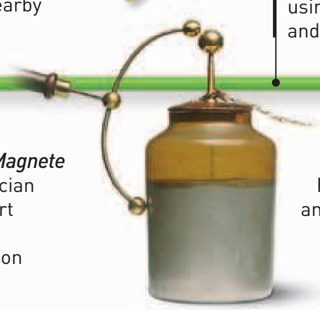


Edison's screw-in light bulb
Although US inventor Thomas Edison is often credited with inventing the light bulb, what he really did was improve an existing idea (see below). He spent years working out a way—using incandescent bulbs—to make electric lighting practical and safe for public use.

600 BCE
Amber
Thales of Miletus rubs a piece of amber against fur and notes that it attracts bits of nearby feathers.


Amber

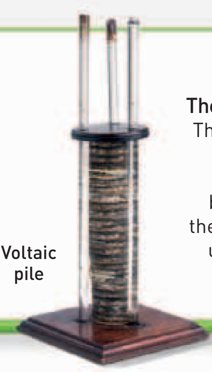
1600
Gilbert's *De Magnete*
English physician William Gilbert publishes his famous work on magnetism.


Leyden Jar

1700–10
Electrostatic generator
English inventor Francis Hauksbee develops a device that can generate static electricity by using a glass globe and wool threads.

1745–46
The Leyden jar
Pieter van Musschenbroe and Ewald Georg von Kleist independently invent a device that allows static electricity to be stored.

1752
Lightning conductor
US scientist Benjamin Franklin flies a kite during a storm, with a key tied to the string, and proves that lightning is a form of electricity.


Voltaic pile

1799–1800
The first voltaic pile
The Italian inventor Alessandro Volta creates the first battery, known as the voltaic pile—the unit "volt" is later named after him.

1820
Electromagnetism discovered
Danish physicist Hans Christian Ørsted notices that a magnet is affected by a nearby wire connected to a battery, and discovers the relationship between magnets and electricity.

1820s–30s
Faraday experiments
English scientist Michael Faraday further illustrates the relationship between electricity and magnetism with his induction ring.


Faraday's induction ring